

COMMENTS OF THE THERMAL BATTERY ALLIANCE

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION
Ensuring the Timely and Orderly Interconnection of Large Loads
Docket No. RM26-4-000
Chairman Laura V. Swett
Federal Energy Regulatory Commission
888 First Street NE
Washington, DC 20426

Dear Chairman Swett and Commissioners:

The Thermal Battery Alliance (TBA) submits these comments in response to the Commission's Advance Notice of Proposed Rulemaking (ANOPR), Ensuring the Timely and Orderly Interconnection of Large Loads. TBA is a coalition of leading U.S. thermal energy storage system manufacturers dedicated to transforming the industrial and power sectors, strengthening U.S. manufacturing, and enhancing grid reliability while lowering energy costs for all. The ANOPR presents a timely opportunity to modernize the regulatory framework governing large-load interconnection in ways that support competitive neutrality, innovation, risk-taking, and system efficiency.

TBA's comments are organized according to the main themes in our policy recommendations, but each section identifies the relevant principles in the Commission's ANOPR. This structure respects the Commission's framework while providing analysis grounded in economic principles and competition-focused perspectives. Throughout these comments, TBA emphasizes that the reforms under consideration can significantly strengthen market performance by enabling flexible loads to participate on fair terms and respond dynamically to price signals. **Allowing large flexible loads the ability to connect expeditiously, manage their wholesale risks and commit to dispatchability will support a more affordable, resilient grid for all customers.**

I. FERC'S AUTHORITY AND THE COMPETITIVE RATIONALE FOR LARGE-LOAD INTERCONNECTION REFORM

ANOPR Principles 1, 2, 3

The ANOPR explains that large-load interconnection falls within the Commission's jurisdiction over transmission service under the Federal Power Act (ANOPR ¶¶ 6–11). TBA agrees that the Commission has authority to establish a framework ensuring that

large-load interconnection is governed by just, reasonable, and non-discriminatory practices.

The jurisdictional line drawn by the ANOPR—providing for FERC jurisdiction over loads and hybrid facilities greater than 20 MW connected to the transmission system—is reasonable. Facilities above this size have grid impacts that require monitoring by the transmission system operator. 20 MW is the threshold used by FERC for large generator interconnection procedures,¹ and such facilities have more rigorous modeling requirements as part of these procedures.²

From a competition perspective, interconnection rules play a central role in market entry conditions. When tariffs treat all loads as inflexible peak contributors, they distort competition, raise barriers to entry, and suppress innovation in flexible industrial technologies. The ANOPR’s call for a modernized, consistent load-interconnection structure is therefore essential not only for reliability but for strengthening competition. By enabling flexible loads to compete on the basis of their actual operational profiles, the Commission reduces regulatory frictions, supports entrepreneurial risk-taking, and improves market efficiency. ERCOT has demonstrated the value of establishing rules that allow for integrated market participation of loads and hybrid facilities. Texas’ Private Use Network rules, enabling participation of “energy parks” which may integrate load, storage, and generation components, has contributed to the region being one of the most competitive locations for new load and hybrid facilities.

II. CHARACTERISTICS AND SYSTEM VALUE OF FLEXIBLE AND HYPER-FLEXIBLE LOADS

ANOPR Principles 4, 5, 14

Flexible, and particularly hyper-flexible loads such as new technologies like thermal energy storage (TES), represent a significant opportunity for enhancing market performance. Long duration, highly flexible storage technologies with high charge to discharge ratios can provide continuous energy output over multi-day periods while only charging during periods of low energy demand and prices, when there is minimal stress on the grid. This operational profile allows these hyper flexible loads to avoid periods of peak scarcity while absorbing energy during low-value hours. In doing so, these hyper flexible and price sensitive loads improve price formation, mitigate congestion, reduce

¹ See *Standardization of Generator Interconnection Agreements & Procs.*, Order No. 2003, 104 FERC 61,103 (2003), *order on reh’g*, Order No. 2003-A, 106 FERC 61,220, *order on reh’g*, Order No. 2003-B, 109 FERC 61,287 (2004), *order on reh’g*, Order No. 2003-C, 111 FERC 61,401 (2005), *aff’d sub nom. Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 475 F.3d 1277 (D.C. Cir. 2007).

² See, e.g. PJM Manual 14G, Attachment B-2 (requiring all inverter-based projects greater than 20 MW to submit a dynamic model).

wasted energy from existing generators, and improve utilization of existing infrastructure, resulting in lower costs for all customers.

The potential scale of such hyper flexible loads is large. For example, by 2030 TES deployment alone will represent many gigawatts of flexible demand over time—meaning that appropriate regulatory treatment is essential to realizing its market and reliability benefits. Notwithstanding recent criticism of the prospects of demand flexibility by certain large loads as merely a “regulatory fiction,”³ hyper-flexible loads like TES are being installed now with the operational capability and business model incentive to make robust, enforceable and dependable flexibility commitments. These loads tend to have low load factors and high price sensitivity, which increases their flexibility. Furthermore, TES can ramp up and down within milliseconds based on price and/or grid operator signals. Such commitments are fundamentally different in nature from commitments to curtail during emergencies made by high load factor customers with a high value of lost load, where such commitments are financial in nature and not directly enforceable by the grid operator. The ANOPR’s framework acknowledges the potential role of flexible loads, but TBA encourages the Commission to more explicitly recognize the unique operational capabilities of hyper-flexible loads as central to modernizing load interconnection.

III. MODELING AND STUDY PRACTICES FOR FLEXIBLE LOADS

ANOPR Principles 6, 7, 8

Accurate modeling is essential for ensuring fair competition. The ANOPR seeks comment on how flexible and curtailable loads should be studied (ANOPR ¶¶ 24–25). TBA strongly recommends that interconnection studies explicitly incorporate enforceable curtailment commitments and multi-day load-shifting capabilities. FERC should require transmission providers to offer a mechanism for load and hybrid facilities to commit to be curtailable and, as applicable, dispatchable, and provide for such commitments to be appropriately accounted for in modeling assumptions.

When a load enforceably commits to avoid peak periods and has the technical capability and durable business model to do so, its interconnection study should not assume peak-period operations. Current modeling practices often inaccurately assume flexible loads will operate inflexibly, resulting in assignment of unnecessary network upgrade costs, resource requirements and reserve margin. This prevents large load customers from moving through the interconnection process and obtaining grid access, even when they are capable of reliable operations using existing system infrastructure.

³ https://www.monitoringanalytics.com/reports/PJM_State_of_the_Market/2025/2025q3-som-pjm.pdf at pp 2-3.

Such mis-modeling functions as a regulatory barrier to competition, encouraging only high-load factor and low-flexibility loads, resulting in an under-utilized grid and higher costs for consumers. When the Commission ensures that modeling reflects actual operational behavior, loads that can and make an enforceable commitment to avoid peak demand will receive cost assignments consistent with cost causation. This strengthens competitive neutrality between flexible and inflexible loads and improves the efficiency of market entry and investment decisions.

IV. FLEXIBLE-LOAD INTERCONNECTION PATHWAYS AND QUEUE REFORM

ANOPR Principles 7, 8, 11

TBA supports the development of interconnection pathways specifically tailored to flexible loads. The Commission should require transmission providers to provide such loads with expedited interconnection treatment by giving them queue priority. In addition, transmission providers should be required to develop streamlined study procedures for such loads so as to facilitate reduced study timelines for loads committing to avoid peaks and follow operator signals for curtailment during reliability events.

Queue reform is essential for reducing entry barriers for innovative industrial technologies. Queue priority is appropriate because loads that can demonstrate asset-backed flexibility pose substantially lower system risks during periods of grid stress and should therefore be eligible for accelerated review.

The Commission should provide clear standards for demonstrating flexibility, including long-term enforceable commitments to respond to RTO instructions, committed operational profiles, and, where appropriate, provisions allowing loads to opt in to direct operator control during system emergencies. Such structures, including faster interconnection for long-term non-firm transmission service with load dispatchability requirements, strengthen incentives for flexible behavior, reduce uncertainty, and support market-based risk management strategies.

V. RTO INFORMATION REQUIREMENTS AND PEAK-AVOIDANCE SIGNALS

Aligned with ANOPR Principles 4, 7

TBA recommends that the Commission require RTOs to provide transparent, mandatory operational signals indicating periods of anticipated congestion, peak demand, or system stress to flexible loads that interconnect via a non-firm transmission service. Flexible loads should be able to rely on these signals to make real-time and day-ahead operational decisions. This is consistent with competitive market design: transparent price and

system-condition information allows market participants to manage their risk exposure, respond to grid needs, and improve overall system performance. RTO signals for mandatory curtailment of flexible loads can help optimize grid operations. Where RTOs are already developing flexible load interconnection and service pathways, FERC should encourage their progress, allowing for later compliance filings should FERC adopt standard requirements.

VI. NON-FIRM SERVICE OPTIONS FOR FLEXIBLE LOADS

ANOPR Principles 6, 7, 8

To facilitate the streamlined interconnection study process described above, and provide for just and reasonable transmission rates, transmission providers should be required to offer non-firm transmission service appropriate for loads that have committed to avoid specific constraints. A non-firm service offering would allow customers to elect the level of firmness that best suits their operational and risk-management preferences. Flexible loads that select non-firm service and can avoid peak loads should receive reduced interconnection and transmission upgrade requirements, and should avoid all network resource and resource adequacy/capacity requirements. Critically, loads and hybrid facilities taking such service should not be obligated to secure commensurate Network Resources to serve the load or hybrid facility, as the non-firm nature of the service allows the customer to be curtailed when Network Resources are not available. Allowing loads to interconnect on a non-firm basis would address a key barrier to timely interconnection for many customers, as the need to secure Network Resources (which may not be available, and even where available add costs) frequently prevents the rapid interconnection of new flexible loads.

This approach aligns risk-bearing with cost responsibility, reinforces market signals, and supports innovation in industrial load management.

VII. CO-LOCATION OF GENERATION AND LOAD

ANOPR Principles 3, 4, 5

TBA strongly encourages the Commission to adopt a clear policy supporting the co-location and adjacent location of generation and flexible loads (forms of dispatchable, curtailable “hybrid facilities,” as that term is used in the ANOPR). Hybrid facilities enable customers to hedge price exposure physically, reduce congestion costs, and mitigate system impacts by reducing gross withdrawals from the grid during peak periods. Under the ANOPR framework (ANOPR ¶¶ 20–23), hybrid configurations should be studied based on net load and the customer’s enforceable commitment to limit or shift consumption to avoid capacity needs. Hybrid facilities are a powerful tool that reduces

barriers to market entry and enhances the ability of firms to manage risk through investment in complementary technologies.

Hybrid facilities may involve different configurations of curtailable load, electric storage and generation resources. For example, a curtailable load such as a TES facility would require energy but not capacity from a co-located or adjacent generation resource. Designation as a hybrid facility should not prevent the generation component of the facility from offering its capacity into the market where the co-located or adjacent load can separately curtail and does not need the capacity.

VIII. COST ALLOCATION AND COST CAUSATION

Aligned with ANOPR Principles 8, 9

Cost allocation should reflect the principle that customers pay the costs they impose on the system, minus any benefits they provide. Flexible loads that avoid peak demand (including based on grid operator signals) do not cause peak-driven upgrades or capacity additions and should not be assigned such costs. When modeling assumptions reflect technical capabilities, realistic load shape forecasts, price sensitivity, enforceable commitments and hybrid facility strategies, cost assignments become more accurate and more consistent with competitive neutrality. This improves the efficiency of investment decisions and ensures that tariff design does not favor legacy technologies over innovative ones.

IX. RELIABILITY AND ENFORCEABLE FLEXIBILITY

ANOPR Principles 10, 11

The ANOPR seeks comment on reliability considerations related to flexible loads (ANOPR ¶¶ 28–30). TBA encourages the Commission to distinguish between enforceable flexibility, such as that provided by TES, and contractual demand response. Enforceable flexibility is flexibility that grid operators can rely on, consistent with the load’s operational capabilities, business model (i.e., value of lost load compared to financial penalties, whether the flexibility is asset-backed, whether the flexibility is an inherent part of the load’s purpose) and, where appropriate, willingness to opt-in to direct operator control during system emergencies. In particular, asset-backed flexibility is more predictable, more reliable and more durable because it arises from technological and operational capabilities rather than administrative arrangements. Recognizing truly enforceable flexibility improves reliability analysis and strengthens the market’s ability to integrate advanced industrial technologies.

X. IMPLEMENTATION AND TRANSITION STRUCTURES

ANOPR Principles 12, 13

TBA supports the development of *pro forma* tariff language establishing flexible-load interconnection service, followed by compliance filings from transmission providers. In the interim, transmission providers should be encouraged to develop tariff mechanisms enabling flexible-load participation even before a final rule is issued. To provide the market with clarity during the transition, the Commission should articulate clear rules for how entities currently proceeding through state-jurisdictional processes will be treated.

We recommend that the Commission state, as soon as feasible, that until FERC has accepted a transmission provider's compliance filing and the transmission provider has begun accepting interconnection requests pursuant to the FERC-jurisdictional process, entities may continue to submit interconnection requests into state-jurisdictional interconnection processes. Further, the Commission should specify that entities that have submitted an interconnection request pursuant to a state-jurisdictional process have the option of continuing with that process, and make clear that interconnection agreements obtained through such a process shall remain valid.

At the same time, customers should have the option to instead pursue the new FERC-jurisdictional pathway (in addition to pathways through any particular RTO). The Commission has authority to set forth transition rules such as these, and has previously taken a similar approach to jurisdictional transitions.⁴ In addition, existing customers should have the option to elect flexible-load non-firm transmission service.

The Commission should also consider a follow-on rulemaking to address issues requiring additional stakeholder analysis. The Commission may provide, for example, for the rapid development of a basic *pro forma* interconnection process, coupled with guidelines on topics that require further development. This would provide for an initial period of flexibility for transmission providers, to be followed by more detailed requirements that phase in once developed by the Commission into fully-implementable *pro forma* language.

XI. WHOLESALE MARKET ACCESS

⁴ See, e.g., *Participation of Distributed Energy Res. Aggregations in Markets Operated by Reg'l Transmission Organizations & Indep. Sys. Operators*, 172 FERC 61,247, 62,661 at PP 101-103 (2020) (Order No. 2222) ("[W]hile we are herein declining to exercise jurisdiction over the interconnections of distributed energy resources to distribution facilities for the purpose of participating in distributed energy resource aggregations, the Commission may revisit this policy decision in the future . . . we recognize that such distributed energy resources may already have interconnected pursuant to procedures that were accepted by the Commission prior to the effective date of this final rule. Therefore, to minimize disruption to existing interconnection agreements for distributed energy resources, we are not requiring distributed energy resources that already interconnected under Commission-jurisdictional procedures to convert to state or local interconnection agreements.").

Aligned with ANOPR Principles 3, 14

FERC rules currently create an uneven playing field for some of the most flexible loads. For example, TES resources that dispatches only electricity will purchase electricity through wholesale transactions; a TES resource that dispatches both heat (to an industrial customer) and electricity (as a wholesale capacity market participant or through a power purchase agreement), would potentially be both a wholesale and retail customer of electricity; a TES resource dispatching only thermal energy would be a retail customer. These three configurations behave similarly or identically on the grid, provide similar value to the grid by increasing energy generation and sales during periods of low demand and prices, and similarly rely on wholesale market access to price follow and avoid peak periods. Furthermore, using TES as a wholesale asset can allow sophisticated industrial customers capable of managing wholesale market risks to access low cost energy to support American reindustrialization. TBA recommends clarifying that wholesale purchases of electricity for conversion into resale as thermal energy qualify as wholesale transactions. Allowing TES to access wholesale markets directly improves competitive neutrality, reduces grid costs, and supports innovation in industrial energy cost management.

XII. CONCLUSION

The ANOPR presents a crucial opportunity to improve competitive market performance, reduce regulatory barriers, and encourage affordable and reliable industrial energy use. By aligning interconnection policy with the operational realities of flexible and hyper-flexible loads, the Commission can improve affordability, support innovation and strengthen the reliability and efficiency of the electric system.

Respectfully submitted,

Isaac Brown

Executive Director

Thermal Battery Alliance